

Notes: Guryan, Hurst, & Kearney (JEP, 2008)

Parental Education and Parental Time with Children

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1 Big picture

- **Question.** How does parental time spent with children vary by education and income, and should economists model child care as home production, leisure, or something distinct?
- **Core empirical fact.** Higher-educated parents spend more time in primary child care, despite having higher opportunity costs of time and despite often working more in the market.
- **Main U.S. data.** The paper uses 2003–2006 American Time Use Survey diaries for parents ages 21–55 with at least one child under 18 and a complete 24-hour diary.
- **Main U.S. finding.** Mothers average about 14.0 hours per week of primary child care; fathers average about 6.8 hours. Mothers with young children average about 21.0 hours.
- **Education gradient.** Women with less than a high school degree spend 12.1 hours per week in total child care, compared with 16.5 hours for college graduates and 17.0 hours for women with more than college.
- **Conditional gradient.** After controlling for age, marital status, number of children, and age of youngest child, the education gradient becomes steeper. Working college-educated women spend about 6 additional hours per week in child care relative to high school dropouts.
- **Not just nonemployment.** The positive gradient holds within working parents and nonworking parents, for mothers and fathers, and across basic, educational, recreational, and travel child care.
- **Contrast with other time uses.** Home production and leisure generally decline with education and income, but primary child care rises with education and income.
- **Primary versus total presence.** More-educated and less-educated parents spend similar total time around children, but more-educated parents spend more time in activities where child care is the primary activity.
- **International finding.** Across a set of time-diary countries, richer countries tend to have higher parental child-care time, and within countries high-educated parents generally spend more child-care time than low-educated parents.
- **Economic takeaway.** Child care is productive like home production and enjoyable like leisure, but the observed gradients imply that parents treat active child-care time as a distinct activity—possibly a luxury good, a high-return human-capital investment, or an input with low substitutability between parental and market care.

2 Data and measurement (time diaries, child care, and education)

2.1 American Time Use Survey sample

- The U.S. analysis uses the 2003–2006 waves of the American Time Use Survey.
- The survey records a 24-hour recall diary for the previous day’s activities.
- Each day of the week is represented, and the paper uses demographic weights adjusted to represent each weekday equally within subgroups.
- The main U.S. sample contains parents ages 21–55 with at least one child under age 18 and a complete 24-hour diary.
- The main Table 1 sample contains 13,434 mothers and 9,259 fathers.

Interpretation. Time diaries are the key measurement device because they avoid relying on broad retrospective guesses about normal weekly time use.

2.2 Definition of total child care

The paper defines total primary child care as

$$\text{PrimaryChildCare}_i = \text{Basic}_i + \text{Educational}_i + \text{Recreational}_i + \text{Travel}_i. \quad (1)$$

- **Basic child care:** feeding, bathing, grooming, medical care, putting a child to sleep, and other basic needs.
- **Educational child care:** reading, teaching, helping with homework, school meetings, and similar activities.
- **Recreational child care:** playing, attending sports or recitals, going to the zoo, taking walks, and similar activities.
- **Travel child care:** travel related to the other child-care categories.

Interpretation. This is a narrow measure of child care. It focuses on activities for which child care is the parent’s primary reported activity, not merely time when a child is present.

2.3 Primary child care versus time around children

The paper emphasizes that child-care measurement can differ depending on whether one measures:

$$\text{PrimaryChildCare}_i \quad \text{or} \quad \text{WithChildren}_i. \quad (2)$$

- Primary child care records activities where caring for children is the main activity.
- The broader “with whom” measure records time when a child is present or accompanying the parent.
- Mothers spend roughly 45 hours per week in the presence of children, and fathers roughly 30 hours.
- These totals are far larger than the 14 and 7 hours of primary child care reported in Table 1.
- The education gradient is strong for primary child care but not for total time around children.

Interpretation. The education gradient is about the nature and intensity of interactions, not simply physical proximity to children.

2.4 Education and income groups

- The U.S. education groups are less than 12 years, 12 years, 13–15 years, 16 years, and more than 16 years.
- In international analysis, “high education” is defined separately within each country to correspond roughly to the upper third of educational attainment among women with children.
- This harmonized definition is necessary because educational systems differ across countries.

Interpretation. Education is used as a proxy for earnings potential and the opportunity cost of time, but also potentially for parental preferences, parenting technology, and beliefs about returns to child investment.

2.5 International time-use sample

- The international analysis uses time-diary surveys from Austria, Canada, Chile, Estonia, France, Germany, Italy, the Netherlands, Norway, Palestine, Slovenia, South Africa, the United Kingdom, and the United States.
- The paper restricts international comparisons to surveys based on 24-hour time diaries.

- For cross-country comparability, the international child-care measure includes basic, educational, and recreational child care, excluding travel.
- Austria and Estonia are excluded from the Table 5 within-country education-gradient regressions because some required controls are unavailable.

Interpretation. The international results are not meant to compare exact hours perfectly across countries. Their strongest evidence comes from within-country education gradients.

3 Models and empirical specifications (equations plus interpretation)

3.1 Beckerian time-allocation problem

A simple way to organize the paper’s logic is a time-allocation model:

$$T_i = \text{MarketWork}_i + \text{HomeProduction}_i + \text{Leisure}_i + \text{PrimaryChildCare}_i + \text{Other}_i. \quad (3)$$

The parent chooses time and market goods to produce utility from consumption, home-produced goods, leisure, and child outcomes:

$$\max U(C_i, H_i, L_i, Q_i) \quad (4)$$

subject to a time constraint and a budget constraint.

Interpretation. If child care were just ordinary home production, higher-wage parents should substitute market services for own time. If child care were just leisure, the pattern would likely resemble leisure. The positive education gradient in primary child care distinguishes it from both.

3.2 Home-production logic

Let child quality be produced by parental time and market inputs:

$$Q_i = q(t_i^p, x_i; A_i), \quad (5)$$

where t_i^p is parental time, x_i is purchased child input, and A_i captures parenting productivity or returns to investment.

Interpretation. A positive education gradient can arise if child quality has high income elasticity, if parental time is hard to substitute with market care, or if higher-educated parents have a higher productivity/return parameter A_i .

3.3 Baseline U.S. education-gradient regression

The conditional differences in Tables 2 and 3 correspond to regressions of the form

$$Y_i = \alpha + \sum_e \beta_e \mathbf{1}\{\text{Education}_i = e\} + f(\text{Age}_i) + \gamma_1 \text{Married}_i + \gamma_2 \text{Children}_i + \gamma_3 \text{AgeYoungest}_i + \varepsilon_i, \quad (6)$$

where the omitted education group is less than 12 years of schooling.

- Y_i can be total child care, market work, nonmarket work, or leisure.
- $f(\text{Age}_i)$ is a cubic in age.
- The coefficients β_e are hours-per-week differences relative to high school dropouts.

Interpretation. A positive β_e for child care and a negative β_e for leisure/home production is the central contrast in the paper.

3.4 International within-country regression

For each country c , the paper estimates an education gradient of the form

$$ChildCare_{ic} = \alpha_c + \beta_c HighEducated_{ic} + \Gamma'_c X_{ic} + u_{ic}, \quad (7)$$

where X_{ic} contains demographic controls such as parental age, marital status, number of children, and age of child when available.

Interpretation. The coefficient β_c asks whether high-educated parents in the same country spend more time in child care than low-educated parents.

3.5 Cross-country income relationship

The paper also describes the cross-country relationship:

$$\overline{ChildCare}_c = a + b \log(GDPpc_c) + \eta_c. \quad (8)$$

Interpretation. The positive relation between GDP per capita and average child-care time supports the view that child care is valued more when income or earnings potential is higher.

3.6 The economic puzzle

The empirical puzzle can be written as

$$\frac{\partial PrimaryChildCare_i}{\partial Education_i} > 0, \quad \frac{\partial HomeProduction_i}{\partial Education_i} < 0, \quad \frac{\partial Leisure_i}{\partial Education_i} < 0. \quad (9)$$

Interpretation. Higher education usually raises the opportunity cost of time, so the positive gradient in child care suggests child-care time is not being chosen like ordinary housework or ordinary leisure.

4 Empirical design and interpretation

4.1 What identifies the main empirical pattern

- The paper is descriptive rather than a causal treatment-effect design.
- The main empirical object is the conditional correlation between parental education/income and time spent with children.
- Controls remove obvious demographic differences such as age, marital status, number of children, and age of youngest child.
- Comparisons are made within working status and gender groups.

Interpretation. The paper does not claim that education causally increases child-care time. It establishes a robust and surprising gradient that theories of time allocation must explain.

4.2 Why the working/nonworking split matters

- Higher-educated parents tend to work more in the labor market.
- If employment were the whole story, higher education should reduce time with children.
- But the education gradient remains positive within working women, nonworking women, and working fathers.

Interpretation. The pattern cannot be dismissed as simply a labor-force participation artifact.

4.3 Why the primary-versus-presence distinction matters

- Parents of all education levels may spend similar time physically around children.
- Higher-educated parents spend more time in activities where child care is the main focus.
- This suggests differences in attention, activity type, or investment intensity.

Interpretation. The mechanism may be about active child investment rather than total supervision.

4.4 Why the international results matter

- The same broad pattern appears outside the United States.
- Cross-country differences show richer countries tend to have higher parental child-care time.
- Within-country regressions show high-educated women generally spend more time in child care than low-educated women.

Interpretation. The education gradient is not an artifact of one U.S. survey or one institutional setting.

4.5 What the paper cannot fully prove

- It cannot separately identify income effects, preference differences, productivity differences, and returns to investment in children.
- It does not measure the quality of child-care time directly.
- It uses primary activity coding, which may undercount supervisory or secondary child care.
- Cross-country comparisons are affected by survey design differences.
- It does not directly estimate child outcomes caused by parental time.

Interpretation. The paper’s strength is documenting a robust empirical regularity; its open question is why the regularity exists.

5 Tables and figures

The paper has no standalone empirical figures in the main text. The key visual material is its sequence of tables. The tables below are directly cropped from the original paper and grouped compactly where feasible.

5.1 Table 1: U.S. child-care time by gender, work status, marital status, and young children

Read:

- All mothers spend 13.96 hours per week in primary child care, while all fathers spend 6.81 hours.
- Basic child care dominates the total: 7.69 hours for mothers and 3.17 hours for fathers.
- Working mothers spend 11.64 hours, still far above working fathers at 6.53 hours.
- Nonworking mothers spend 18.68 hours, about 7 more hours than working mothers.

- Mothers with children under age 5 spend 21.03 hours per week; fathers with children under age 5 spend 9.69 hours.

Economic message: Gender and employment matter, but even working mothers spend substantially more child-care time than working fathers. Young children sharply raise primary child-care time.

Table 1

Hours per Week Spent in Child Care by Various Subgroups within the United States: 2003–2006

	<i>All child care</i>	<i>Basic child care</i>	<i>Educational child care</i>	<i>Recreational child care</i>	<i>Travel child care</i>
All mothers (<i>n</i> = 13,434)	13.96	7.69	2.10	2.27	1.90
All fathers (<i>n</i> = 9,259)	6.81	3.17	0.88	1.83	0.93
Working mothers (<i>n</i> = 9,372)	11.64	6.40	1.72	1.66	1.86
Working fathers (<i>n</i> = 8,553)	6.53	3.05	0.82	1.76	0.91
Nonworking mothers (<i>n</i> = 4,062)	18.68	10.32	2.89	3.50	1.97
Nonworking fathers (<i>n</i> = 706)	9.90	4.65	1.55	2.54	1.15
Married working mothers (<i>n</i> = 6,323)	12.00	6.64	1.74	1.78	1.84
Married working fathers (<i>n</i> = 7,559)	6.74	3.17	0.82	1.85	0.90
Married nonworking mothers (<i>n</i> = 3,065)	19.56	10.73	2.97	3.75	2.11
Married nonworking fathers (<i>n</i> = 529)	10.59	5.27	1.64	2.46	1.22
Unmarried working mothers (<i>n</i> = 3,049)	10.76	5.80	1.66	1.39	1.91
Unmarried working fathers (<i>n</i> = 994)	4.91	2.10	0.75	1.04	1.02
Unmarried nonworking mothers (<i>n</i> = 997)	15.72	8.87	2.72	2.58	1.56
Unmarried nonworking fathers (<i>n</i> = 177)	8.03	3.09	1.50	2.51	0.93
Mothers with children under 5 (<i>n</i> = 5,220)	21.03	12.62	1.73	4.75	1.93
Fathers with children under 5 (<i>n</i> = 3,900)	9.69	4.77	0.75	3.31	0.85

Table 1: Hours per week spent in child care by U.S. subgroups

5.2 Table 2: U.S. child-care time for women by education

Read:

- Women with less than 12 years of education spend 12.1 hours per week in total child care.
- Women with 16 years of education spend 16.5 hours, and women with more than 16 years spend 17.0 hours.
- Highly educated women work more in the market and have fewer children, making the positive child-care gradient more striking.
- Conditional differences are larger than raw differences: among nonworking women, college graduates spend 6.4 extra hours and women with more than college spend 9.7 extra hours relative to high school dropouts.
- Among working women, the corresponding conditional differences are 6.1 and 6.4 hours.

Economic message: Higher education raises market work but also raises active child-care time. This is the central empirical puzzle.

Table 2

Hours per Week Spent in Total Child Care for Women in the United States by Educational Attainment

Years of schooling	Fraction married	Fraction working	Total market work	Number of children	Hours per week spent in total child care				
					All	Nonworking		Working	
						Conditional (relative to education < 12)		Conditional (relative to education < 12)	
<12	0.63	0.42	14.6	2.2	12.1	14.9		8.3	
12	0.69	0.65	22.3	1.9	12.6	17.6	2.9	9.8	3.1
13–15	0.69	0.74	25.3	1.9	13.3	18.9	3.9	11.4	4.2
16	0.87	0.72	23.7	1.9	16.5	22.6	6.4	14.2	6.1
16+	0.89	0.79	27.4	1.8	17.0	25.9	9.7	14.4	6.4

Table 2: Total child-care time for women by educational attainment**5.3 Table 3: Education gradients in child care, market work, home production, and leisure****Read:**

- For women, the child-care gradient is positive in both nonworking and working samples.
- Nonworking women with more than college spend 9.70 more hours in child care relative to high school dropouts.
- Working women with more than college spend 6.36 more hours in child care relative to high school dropouts.
- Leisure falls sharply with education. Working men with more than college spend 7.96 fewer hours in leisure relative to high school dropouts.
- Nonmarket work also generally falls with education among women.

Economic message: Child care moves in the opposite direction from home production and leisure. That is why the authors argue child care should be modeled as distinct.

Table 3

Conditional Differences in Hours per Week Spent in Different Time Use Categories By Education: Relative to High School Dropouts

Years of schooling	Nonworking			Working			
	Total child care	Total nonmarket work	Leisure	Total child care	Total market work	Total nonmarket work	Leisure
Panel A: Women (relative to high school dropouts)							
12	2.85 (0.88)	-1.35 (1.15)	-3.71 (1.39)	3.09 (0.62)	-0.51 (1.83)	-2.19 (1.01)	-0.71 (1.44)
13–15	3.88 (0.87)	-1.65 (1.16)	-7.30 (1.42)	4.24 (0.61)	0.35 (1.81)	-2.45 (1.00)	-3.49 (1.41)
16	6.33 (1.08)	-3.08 (1.28)	-8.05 (1.54)	6.07 (0.64)	-0.27 (1.89)	-2.94 (1.06)	-4.39 (1.45)
16+	9.70 (1.64)	-6.08 (1.63)	-7.29 (1.88)	6.36 (0.73)	1.67 (2.04)	-4.41 (1.13)	-4.75 (1.53)
Panel B: Men (relative to high school dropouts)							
12	-2.57 (1.67)	-1.88 (3.03)	6.33 (3.57)	2.38 (0.42)	0.70 (1.64)	-0.53 (0.79)	-3.61 (1.40)
13–15	-0.13 (1.77)	2.37 (2.98)	-8.94 (3.83)	3.32 (0.46)	-0.16 (1.66)	1.09 (0.81)	-6.41 (1.38)
16	3.89 (4.48)	5.48 (3.96)	-10.38 (4.75)	4.27 (0.43)	0.81 (1.68)	0.36 (0.83)	-7.82 (1.37)
16+	-1.24 (2.57)	5.48 (3.96)	-10.37 (5.64)	4.76 (0.47)	2.16 (1.81)	-1.11 (0.85)	-7.96 (1.49)

Table 3: Conditional education gradients across time-use categories

5.4 Table 4: International child-care time ranked by GDP per capita

Read:

- Norway and the United States are near the top, with about 8.9 hours per week for all adults with children.
- U.S. women average 11.64 hours and U.S. men average 5.62 hours using the internationally harmonized measure.
- Canada, Austria, Norway, and the United States show high parental child-care time.
- South Africa and France are low in the table for average child-care time.
- Gender gaps vary dramatically across countries; South Africa has a very large mother-father gap.

Economic message: The positive income pattern is not only within the United States. Richer countries tend to spend more parental time in child care.

Table 4
Hours per Week Spent in Child Care across Countries: Ranked by GDP per Capita

<i>Country</i>	<i>GDP per capita</i>	<i>All with children</i>	<i>Men with children</i>	<i>Women with children</i>
Norway	37,200	8.88	5.68	11.70
United States	34,300	8.93	5.62	11.64
United Kingdom	24,500	7.23	4.17	9.83
Netherlands	24,200	6.83	4.36	8.91
Austria	23,900	8.33	3.57	12.27
Canada	23,600	8.56	5.61	11.20
France	21,800	4.43	1.82	6.83
Germany	23,100	7.25	3.87	10.49
Italy	19,000	7.32	4.03	10.37
Slovenia	9,700	5.03	2.75	7.21
Chile	4,900	5.65	2.69	7.90
Estonia	4,100	6.64	2.97	9.37
South Africa	2,900	4.03	0.60	5.88
Palestine	1,200	6.38	2.09	12.35

Table 4: Child-care time across countries

5.5 Table 5: International education gradients among women

Read:

- The table reports within-country differences between high-educated and low-educated women.
- Working high-educated women spend more child-care time in every listed country; the U.S. difference is 3.39 hours per week.
- Nonworking high-educated women also generally spend more child-care time; the U.S. difference is 5.43 hours.
- The main exception is nonworking women in Chile, where the difference is approximately zero.
- The United Kingdom, Norway, Palestine, Slovenia, Canada, and the United States all show notable positive differences among nonworking women.

Economic message: The education gradient is not uniquely American. Within many different countries, high-educated mothers devote more active time to children.

Table 5

Conditional Differences in Hours per Week Spent in Total Child Care for Women by Educational Attainment

Country	Working women with kids		Nonworking women with kids	
	Fraction high educated	Difference in time spent on child care by high educated	Fraction high educated	Difference in time spent on child care by high educated
Canada	0.36	1.21 (0.47)	0.29	2.91 (1.35)
Chile	0.47	2.18 (1.54)	0.23	-0.11 (1.21)
France	0.37	0.47 (0.35)	0.24	1.47 (0.70)
Germany	0.35	0.04 (0.34)	0.25	0.81 (0.84)
Italy	0.59	2.06 (0.44)	0.33	1.86 (0.62)
Netherlands	0.39	2.01 (1.31)	0.24	0.25 (1.33)
Norway	0.28	1.15 (0.64)	0.15	6.31 (2.06)
Palestine	0.63	1.56 (0.95)	0.27	2.89 (0.86)
Slovenia	0.31	1.52 (0.75)	0.14	2.31 (2.47)
South Africa	0.33	0.58 (0.66)	0.21	0.99 (1.12)
United Kingdom	0.35	1.87 (0.56)	0.22	4.17 (1.77)
United States	0.31	3.39 (0.33)	0.22	5.43 (0.77)

Table 5: International education gradients in child-care time for women

6 Short summary

- The paper documents how parental child-care time varies by education, income, gender, work status, and country.
- The main U.S. data source is the 2003–2006 American Time Use Survey.
- The main outcome is primary child-care time, broken into basic, educational, recreational, and travel child care.
- Mothers spend about twice as much primary child-care time as fathers.
- Parents with young children spend much more time in child care than parents with older children.
- Higher-educated mothers spend more time in child care even though they are more likely to work and have fewer children.
- Conditional on demographics, college-educated working women spend roughly 6 extra hours per week in child care relative to women with less than high school.
- The positive education gradient appears for both working and nonworking women and for working fathers.
- Primary child-care time rises with education, but home production and leisure generally fall with education.
- Parents of different education levels spend similar total time around children, but high-educated parents spend more time in child-focused primary activities.
- Across countries, higher GDP per capita is associated with more parental child-care time.
- Within countries, high-educated women generally spend more time in child care than low-educated women.

- The authors conclude that child care should not be treated as simply ordinary home production or ordinary leisure.
- Possible explanations include child care being a luxury good, parental care being hard to substitute with market care, higher returns to investing in high-education households' children, or stronger preferences for active child investment.
- The result matters for intergenerational mobility because children of higher-educated parents receive more active parental time.

7 Conclusion

7.1 Core conclusion

Guryan, Hurst, and Kearney show that more-educated and higher-income parents spend more time in primary child care, even though their opportunity cost of time is higher. This finding is robust across U.S. demographic subgroups and appears in international time-diary data as well.

7.2 Economic intuition

The result is surprising under a simple time-allocation model. Higher wages usually lead people to economize on time-intensive home production and leisure. Yet parental child-care time rises with education. This means active child care is likely chosen for reasons that go beyond ordinary household production: it may be an investment in children's human capital, a high-value consumption good, or an activity with low substitutability between parental and market inputs.

7.3 Takeaway

The paper's lasting message is that parental time is an economically important input into child development and the transmission of human capital. Because more-educated parents spend more active time with children, parental time may reinforce intergenerational inequality even before differences in schools, neighborhoods, or financial resources are considered.